

FILTER DESIGN TEST CASE

Coupled Resonator Filter - Band 25 (1900 MHz) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0073
Revision:	Rev B.2
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Created Date:	2024-08-23
Last Modified:	2024-10-08
Status:	CONDITIONAL PASS
Priority:	P4 - Low

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting Band 25 (1900 MHz) for Public Safety Band Filter. The filter is designed on AlN on Si substrate with a target center frequency of 3511.9 MHz and bandwidth of 76.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: Band 25 (1900 MHz)
- Application: Public Safety Band Filter
- Substrate: AlN on Si
- Package: Flip-Chip BGA
- Design Tool: CST Studio Suite 2024
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Coupled Resonator Filter design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3511.9 MHz
Bandwidth (3 dB):	76.8 MHz
Insertion Loss Target:	≤ 1.2 dB
Return Loss Target:	≥ 9.0 dB
Out-of-Band Rejection:	≥ 32.0 dB
Isolation (Duplexer):	≥ 18.0 dB
Q Factor:	13626
Temperature Coefficient:	-22.0 ppm/°C
Die Size:	0.7 x 1.2 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the Coupled Resonator Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3511.9 MHz, BW 76.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10536 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 115 MHz offset >= 32.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 9.7 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2024-10-08
Devices Tested: 30
Insertion Loss (meas): 0.86 dB
Return Loss (meas): 11.3 dB
Rejection (meas): 34.7 dB
Isolation (meas): 22.9 dB
Q Factor (meas): 13626
Group Delay Variation: 1.2 ns
Power Handling: +29 dBm
Die Yield: 92.6%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated acceptable performance for Band 25 (1900 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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